

AHMAD MARAQA

AI Research Engineer | AI/ML · LLM Agents · Computer Vision · AI Automation

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[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFILE

AI Research Engineer working across AI/ML, large language models, computer vision and AI automation. I train and evaluate models on real, noisy data, build LLM systems on RAG, embeddings and tool-calling agents, and run the automation that connects them to databases and messaging platforms in production.

CORE SKILLS

- AI & Machine Learning** Classification · Regression · Multi-Label Classification · Ensemble Methods · Model Evaluation · Deep Learning · Neural Networks · CNNs · TensorFlow · Keras · Scikit-learn · Pandas · NumPy
- LLMs & Agents** RAG · Embeddings · Vector Databases (Pinecone, FAISS) · Prompt Engineering · Tool Calling · Structured Output · Local & Open-Source LLMs · Ollama · LangChain · LlamaIndex · OpenRouter · Groq · Claude Code
- Computer Vision** Object Detection · Image Classification · YOLOv8 · OpenCV · Ultralytics · CVAT · Data Annotation · Data Augmentation
- Automation & APIs** Python · n8n (incl. self-hosted) · MongoDB · WhatsApp Business Cloud API · Telegram Bot API · Google Workspace APIs · REST APIs · Webhooks · Webhook Security
- Deployment & Ops** Hostinger VPS (self-hosted n8n, Chatwoot) · Cloudflare · Domains & DNS · Streamlit · GitHub Pages · Git · IoT · BLE

PROFESSIONAL EXPERIENCE

- AI Research & Integration Engineer** — **Penguin Location Services** Jan 2025 – Present
- Research and implement AI/ML solutions for real-world applications, including machine learning models for IoT and RTLS environments.
 - Develop LLM-based applications using RAG, embeddings, vector databases, prompt engineering and local/open-source LLMs via Ollama.
 - Build AI-powered assistants and integrations connecting language models with external services and tools.
- AI Research Engineer Intern** — **Penguin Location Services** Oct 2024 – Jan 2025
- Trained and evaluated KNN, Decision Tree and Random Forest models on BLE sensor data, and implemented a custom probabilistic model keeping prediction confidence in the output.
- AI & ML Engineer Intern** — **Leading Point Software** Jul 2024 – Sep 2024
- Built machine learning and computer vision solutions including image classification and object detection; hands-on with AutoML tools.

INDEPENDENT WORK — ONGOING

- AI Automation & Agents — Independent** — **Self-taught** 2025 – Present
- Develop and maintain production automation workflows using n8n, MongoDB, APIs and webhooks — 30+ built to date, learned independently outside of any role.
 - Integrate the WhatsApp Business Cloud API into automated workflows and AI-powered applications: approved templates, verified webhooks, rate-limit-aware batching, and timed maintenance jobs.
 - Build tool-calling AI agents on the Telegram Bot API — task management, scheduling, voice in and out, document review, and customer-facing support with human escalation — integrated with Google Sheets, Calendar, Gmail and RAG over Pinecone.
 - Self-host and maintain n8n and Chatwoot on a Hostinger VPS, including credentials, HTTPS, persistence, domains, DNS and Cloudflare.

SELECTED PROJECTS

- Jordanian Currency Detection & Counting** — YOLOv8 detection on a self-collected, augmented dataset; no public one existed. Deployed as a web app. [\[Live\]](#) [\[Code\]](#)
- Fire Detection from IoT Sensors** — Binary classification on ~62k tabular readings, 13 features; seven models compared side by side. [\[Live\]](#) [\[Code\]](#)
- Rejecting Unknown Classes in Classification** — Same CNN twice, changing only the output layer and loss, so it can decline inputs outside its classes. [\[Notebook\]](#)
- Dublejli — AI Video Dubbing System** — Speech-to-text, translation and text-to-speech with video sync. Exhibited at SOFEX 2024 and NTP 2024.

EDUCATION

- B.Sc. Computer Engineering** — **University of Jordan** Oct 2020 – Sep 2024
- GPA 3.77 / 4.00 — Excellent • Languages: Arabic (native), English (professional working proficiency)